

YGGDRASIL — TRUTH-ALIGNED BUILD

A Kernel-Level DNS Defense Architecture — Explicit Status Classification

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Abstract

This document presents Yggdrasil as a compositional architecture. All claims are explicitly labeled as IMPLEMENTED, PROTOTYPE, PROPOSED, or SPECULATIVE. No claim exceeds available support.

1. Problem Space [REAL]

DNS lacks inherent verification, privacy, and memory. Cache poisoning, plaintext observation, and routing attacks are well-documented realities.

2. System Architecture [PROPOSED]

$Y = (I, V, S, M, F)$ where Interception, Verification, Shadow, Memory, Federation are composed at a kernel boundary.

3. Kernel Interception [PROTOTYPE]

eBPF-based interception is feasible and partially implemented in modern systems. Full production reliability and coverage are not established here.

4. Verification Pipeline [PROPOSED]

DNSSEC validation, source validation, and anomaly scoring are described but not fully implemented or benchmarked.

5. Shadow Plane [PROPOSED]

Deception-based routing of suspicious traffic is conceptually valid but not demonstrated at scale in this work.

6. Memory System [PROTOTYPE]

Event logging and persistence concepts are valid. Weighted adaptive learning is not empirically validated here.

7. Federation [SPECULATIVE]

Peer-to-peer sharing of threat intelligence is proposed but lacks Byzantine fault tolerance and validation.

8. Performance [UNVERIFIED]

Latency estimates are theoretical and not backed by measurement in this document.

9. Legal Considerations [REAL / UNRESOLVED]

Deception-based defense introduces jurisdictional ambiguity and is not resolved here.

Conclusion

Yggdrasil is a structured architectural proposal grounded in real problem space but not yet proven as a production system.